

main.c	Run	Output
<pre>1 #include <stdio.h> 2 #include <stdlib.h> 3 int main() 4 { 5 int r[50], n, head, size; 6 int i, j, temp, total = 0; 7 printf("Enter number of disk requests: "); 8 scanf("%d", &n); 9 printf("Enter disk request queue: "); 10 for(i = 0; i < n; i++) 11 scanf("%d", &r[i]); 12 printf("Enter initial head position: "); 13 scanf("%d", &head); 14 printf("Enter disk size: "); 15 scanf("%d", &size); 16 for(i = 0; i < n - 1; i++) 17 for(j = i + 1; j < n; j++) 18 if(r[i] > r[j]) 19 { 20 temp = r[i]; 21 r[i] = r[j]; 22 r[j] = temp; 23 } 24 printf("\nSeek Sequence:\n%d", head); 25 for(i = 0; i < n; i++) 26 { 27 if(r[i] >= head) 28 { 29 total += abs(head - r[i]); 30 head = r[i]; 31 printf(" -> %d", head); 32 } 33 } 34 total += abs(head - (size - 1)); 35 head = size - 1; 36 printf(" -> %d", head); 37 total += size - 1; 38 head = 0; 39 printf(" -> %d", head); 40 for(i = 0; i < n; i++) 41 { 42 if(r[i] < head) 43 { 44 total += abs(head - r[i]); 45 head = r[i]; 46 printf(" -> %d", head); 47 } 48 } 49 printf("\n\nTotal Head Movement = %d cylinders\n", total); 50 printf("Average Head Movement = %.2f cylinders\n", 51 (float)total / n); 52 return 0; 53 }</pre>		<pre>Enter number of disk requests: 6 Enter disk request queue: 20 35 70 90 10 50 Enter initial head position: 45 Enter disk size: 100 Seek Sequence: 45 -> 50 -> 70 -> 90 -> 99 -> 0 Total Head Movement = 153 cylinders Average Head Movement = 25.50 cylinders === Code Execution Successful ===</pre>